

Boat Monitor P2 — Wiring (Minimal Changes)

Keep sensors + VSNS + LEDs · RST on pin 24 · Key on pin 15

SECTION 1 — ALREADY DONE (do not change)

Pin	Goes to
1, 2	Engine INA260 SDA, SCL
4, 5	House INA260 SDA, SCL
6, 7	INA219 SDA, SCL
9	TPS STAT
16	TPS VSNS — KEEP (firmware Pin 12)
17, 19, 20	Red, Blue, Green LED — KEEP
39	TPS OUT → VSYS

VSNS pin 16 works. Old bug was code Pin(16)=GPIO16=pin 21, not silkscreen 16. Use Pin(12) in firmware.

Do NOT change INA260 A0/A1 pads or I2C wires.

SECTION 2 — STILL TO DO

SIM7600 (jumpers: UART=B, PWR=PWR-3V3, VCCIO=3.3V)

Modem	Connect to
5V	5V rail
GND	GND rail
RXD	Pico pin 11
TXD	Pico pin 12
RST	Pico pin 24 (NOT 19 — LED stays on 19)

Optocoupler — INPUT side (12V boat)

Opto IN	Boat wire
Ch1 IN+	Mid bilgde
Ch2 IN+	Aft bilgde
Ch3 IN+	Mid water return
Ch4 IN+	Aft water return
Ch5 IN+	Switch (also → PlusRoc IN+)
Ch6 IN+	Key
All IN-	Ground bus

Optocoupler — OUTPUT side (3.3V Pico)

Opto OUT	Pico pin	Signal
OUT1	21	Mid bilge pump
OUT2	22	Aft bilge pump
OUT3	25	Mid float
OUT4	10	Aft float
OUT5	14	Switch sense
OUT6	15	Key sense

VCC→3V3 rail, GND→GND, jumpers=pull-up. RUN: OUT1-4 → 10kΩ → pin 30

PlusRoc

PlusRoc	Connect to
IN+	Switch boat wire
OUT+ 5V	TPS IN2 + V50 charge + modem 5V
GND	Ground bus

Switch vs Key

Switch = master battery ON (powers boat). Key = ignition ON even if engine not charging.

Checklist

☐	Task
☐	Modem 5V GND pins 11 12 24
☐	Opto + RUN 10k Ω
☐	PlusRoc
☐	16 boat wires (page 2)

SECTION 3 — Final pin map

Pin	Status	Goes to
1-7, 9, 16-20	DONE	I2C, STAT, VSNS, LEDs
10, 14-15	TODO	Opto OUT4,5,6
11-12	TODO	Modem UART
21-22, 25	TODO	Opto OUT1,2,3
24	TODO	Modem RST
30	TODO	RUN
39	DONE	VSYS

SECTION 4 — 16 boat wires

#	Label	Connect to
1	Engine +	Bottom INA260 OUT+
2	Engine -	Bottom GND + ground bus
3	House +	Middle INA260 OUT+
4	House -	Middle GND + ground bus
5	Mid bildge	Opto Ch1 IN+
6	Aft bilgde	Opto Ch2 IN+
7	Key	Opto Ch6 IN+
8	Switch	PlusRoc IN+ + Opto Ch5 IN+
9	Mid water	12V bus
10	Mid water	Opto Ch3 IN+
11	Solar engin	Bottom INA260 IN+
12	Solar engin	Bottom OUT+
13	Solar hous	Middle INA260 IN+
14	Solar hous	Middle OUT+
15	Aft water	12V bus
16	Aft water	Opto Ch4 IN+